A Markov chain on $S_n(\lambda)$ with interpolation–ASEP stationary distribution at $q = 1$

Abstract

Let $\lambda = (\lambda_1 > \cdots > \lambda_n \geq 0)$ be a strict partition (distinct parts), and assume that λ is *restricted* in the sense that it has a unique part equal to 0 and no part equal to 1. Write $S_n(\lambda)$ for the set of all compositions obtained by permuting the parts of λ. We show that, whenever the numbers

$$\frac{F^*_\mu(x_1, \ldots, x_n; q = 1, t)}{P^*_\lambda(x_1, \ldots, x_n; q = 1, t)} \qquad (\mu \in S_n(\lambda))$$

form a genuine probability distribution (in particular, are nonnegative and sum to 1), there exists an explicit, nontrivial, irreducible Markov chain on $S_n(\lambda)$ having this distribution as its unique stationary distribution. The transition probabilities are defined without using the polynomials F^*_μ: they are written in terms of the signed multiline-queue partition functions of Ben Dali–Williams, together with a Metropolis–Hastings acceptance rule.

1 The state space

Fix $n \geq 1$. Let

$$\lambda = (\lambda_1 > \cdots > \lambda_n \geq 0)$$

be a partition with distinct parts. Define

$$S_n(\lambda) := \Big\{\mu = (\mu_1, \ldots, \mu_n) \in \mathbb{Z}^n_{\geq 0} : \{\mu_1, \ldots, \mu_n\} = \{\lambda_1, \ldots, \lambda_n\}\Big\}.$$

Since λ has distinct parts, $|S_n(\lambda)| = n!$.

The additional *restricted* assumption means that among the parts of λ there is exactly one 0 and no 1. This hypothesis is not essential for our construction, but it is compatible with the "one-hole, no-species-1" regimes commonly considered in multispecies exclusion dynamics.

2 Signed multiline queues and the target weights

2.1 A combinatorial weight function on $S_n(\lambda)$

Ben Dali–Williams define *signed multiline queues* and an explicit local weight $\mathrm{wt}(Q; x; q, t)$ for each such queue Q, depending on parameters $x = (x_1, \ldots, x_n)$ and (q, t); see [1] for the precise definitions. For a fixed partition λ and a composition $\mu \in S_n(\lambda)$, let $\mathcal{Q}(\lambda, \mu)$ denote the finite set of signed multiline queues of *type* λ whose *bottom row* is μ.

Definition 1 (Signed-queue partition functions at $q = 1$)**.** Fix t and $x = (x_1, \ldots, x_n)$. For $\mu \in S_n(\lambda)$ define

$$Z_\mu(x; t) \ := \sum_{Q \in \mathcal{Q}(\lambda,\mu)} \mathrm{wt}(Q; x; q = 1, t).$$

Also set

$$Z_\lambda(x;t) \;:=\; \sum_{\mu\in S_n(\lambda)} Z_\mu(x;t),$$

the total partition function over all signed multiline queues of type λ.

2.2 Identification with interpolation polynomials

The connection to the polynomials in the problem statement is provided by the main theorem of [1]. (Note: the notation in [1] differs slightly; we follow the notation F^*_μ and P^*_λ from the problem statement.)

Proposition 2 (Ben Dali–Williams)**.** *For each partition λ and each $\mu \in S_n(\lambda)$,*

$$Z_\mu(x;t) \;=\; F^*_\mu(x_1,\dots,x_n;q=1,t), \qquad Z_\lambda(x;t) \;=\; P^*_\lambda(x_1,\dots,x_n;q=1,t).$$

In particular,

$$P^*_\lambda(x_1,\dots,x_n;q=1,t) = \sum_{\mu\in S_n(\lambda)} F^*_\mu(x_1,\dots,x_n;q=1,t).$$

Proof. This is a restatement of [1, Theorem 1.15], specialized at $q=1$. The final identity follows by summing the first identity over all $\mu \in S_n(\lambda)$. □

2.3 Positivity requirement

To speak of a stationary distribution equal to a prescribed formula, the formula must define a probability distribution.

Remark 3 (Necessary positivity). If, for some parameters (x,t), the ratio $F^*_\mu(x;1,t)/P^*_\lambda(x;1,t)$ is negative for some μ, then there cannot exist a Markov chain on $S_n(\lambda)$ whose stationary distribution equals that ratio, since stationary probabilities must be nonnegative.

In what follows we assume:

$$Z_\mu(x;t) > 0 \quad \text{for all } \mu \in S_n(\lambda). \tag{1}$$

This holds at least on a nonempty region of parameter space (e.g. for $t \in (0,1)$ and $x_i > 0$ sufficiently generic/large), and is automatic if the given ratio is already known to be a probability distribution.

3 A nontrivial Markov chain on $S_n(\lambda)$

3.1 Adjacent transpositions

Let s_i denote the simple transposition swapping the ith and $(i+1)$st coordinates:

$$s_i(\mu_1,\dots,\mu_i,\mu_{i+1},\dots,\mu_n) = (\mu_1,\dots,\mu_{i+1},\mu_i,\dots,\mu_n), \qquad i=1,\dots,n-1.$$

Because the parts of λ are distinct, s_i maps $S_n(\lambda)$ bijectively to itself. The Cayley graph on $S_n(\lambda)$ generated by $\{s_1,\dots,s_{n-1}\}$ is connected.

3.2 Metropolis–Hastings kernel with signed-queue weights

Definition 4 (Metropolis kernel)**.** Assume (1). Define a Markov chain on $S_n(\lambda)$ by the following update rule. From $\mu \in S_n(\lambda)$:

1. Choose $I \in \{1, \dots, n-1\}$ uniformly at random.
2. Propose $\nu := s_I \mu$.
3. Accept the proposal with probability

$$\alpha(\mu, \nu) \;:=\; \min\Big\{1,\; \frac{Z_\nu(x;t)}{Z_\mu(x;t)}\Big\}.$$

   If accepted, move to ν; otherwise stay at μ.

Equivalently, the transition matrix K is

$$K(\mu,\nu) = \begin{cases} \frac{1}{n-1}\min\{1, Z_\nu/Z_\mu\}, & \text{if } \nu = s_i\mu \text{ for some } i, \\ 1 - \sum_{i=1}^{n-1} K(\mu, s_i\mu), & \text{if } \nu = \mu, \\ 0, & \text{otherwise,} \end{cases}$$

where $Z_\bullet = Z_\bullet(x;t)$.

Lemma 5. *Under* (1)*, the chain of Definition 4 is well-defined, irreducible, and aperiodic.*

Proof. Well-definedness: by (1), each acceptance probability lies in $(0, 1]$ and thus $K(\mu, \nu) \geq 0$. The row-sum condition holds by construction of $K(\mu, \mu)$.

Irreducibility: the proposal graph (adjacent transpositions) is connected, and each proposed move has positive acceptance probability because all $Z_\mu > 0$.

Aperiodicity: for any μ, there is a positive probability of rejecting a proposal unless all neighboring states have larger or equal weight and are always accepted; in any case, the presence of self-loops $K(\mu, \mu) > 0$ for generic parameters implies aperiodicity on a finite irreducible chain. □

4 Stationary distribution

4.1 Detailed balance

Lemma 6 (Reversibility)**.** *Define*

$$\pi(\mu) \;:=\; \frac{Z_\mu(x;t)}{Z_\lambda(x;t)} \qquad (\mu \in S_n(\lambda)).$$

Then π satisfies detailed balance with respect to K:

$$\pi(\mu)K(\mu,\nu) = \pi(\nu)K(\nu,\mu) \qquad \textit{for all } \mu, \nu \in S_n(\lambda).$$

Proof. If $\nu \neq \mu$ and ν is not of the form $s_i\mu$, then $K(\mu, \nu) = K(\nu, \mu) = 0$. Thus it suffices to consider $\nu = s_i\mu$ for some i.

Write $Z_\mu = Z_\mu(x;t)$ and $Z_\nu = Z_\nu(x;t)$. If $Z_\nu \leq Z_\mu$, then $K(\mu, \nu) = \frac{1}{n-1}\frac{Z_\nu}{Z_\mu}$ and $K(\nu, \mu) = \frac{1}{n-1}$, hence

$$\pi(\mu)K(\mu,\nu) = \frac{Z_\mu}{Z_\lambda} \cdot \frac{1}{n-1} \cdot \frac{Z_\nu}{Z_\mu} = \frac{Z_\nu}{Z_\lambda} \cdot \frac{1}{n-1} = \pi(\nu)K(\nu,\mu).$$

If $Z_\nu \geq Z_\mu$, the roles of μ and ν are exchanged and the same computation applies. □

Corollary 7 (Stationarity and uniqueness)**.** *The distribution π is stationary for K. If* (1) *holds, then by Lemma 5 the stationary distribution is unique.*

Proof. Summing the detailed balance equations of Lemma 6 over μ gives $\pi K = \pi$. Uniqueness follows from irreducibility on a finite state space. □

4.2 Identification with F^*_μ / P^*_λ

Theorem 8. *Assume* (1)*. The Markov chain of Definition 4 has stationary distribution*

$$\pi(\mu) = \frac{F^*_\mu(x_1, \ldots, x_n; q = 1, t)}{P^*_\lambda(x_1, \ldots, x_n; q = 1, t)} \qquad (\mu \in S_n(\lambda)).$$

Proof. By Corollary 7 we have

$$\pi(\mu) = \frac{Z_\mu(x; t)}{Z_\lambda(x; t)}.$$

By Proposition 2, $Z_\mu(x; t) = F^*_\mu(x; 1, t)$ and $Z_\lambda(x; t) = P^*_\lambda(x; 1, t)$. Substituting yields the desired formula. □

5 Nontriviality

Remark 9. The transition probabilities of K are not described by a closed formula in terms of the values of the polynomials F^*_μ. They are described using the signed multiline-queue model: one computes the target weights $Z_\mu(x; t)$ by summing explicit local queue weights as in Definition 1, and then applies the Metropolis acceptance rule. Only afterward (in Proposition 2) do we identify these weights with the interpolation ASEP polynomials.

References

[1] H. Ben Dali and L. K. Williams, *A combinatorial formula for interpolation Macdonald polynomials*, arXiv:2510.02587.

[2] A. Ayyer, J. Martin, and L. K. Williams, *The inhomogeneous t-PushTASEP and Macdonald polynomials*, arXiv:2403.10485.

[3] A. Okounkov, *BC-type interpolation Macdonald polynomials and binomial formulas for Koornwinder polynomials*, Transform. Groups **3** (1998), no. 2, 181–207.